Lab 7: Fuzzing for Input Validation Bugs

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Program: Pet Walking Service (processor.py)

Bugs found by Hypothesis and how they were fixed

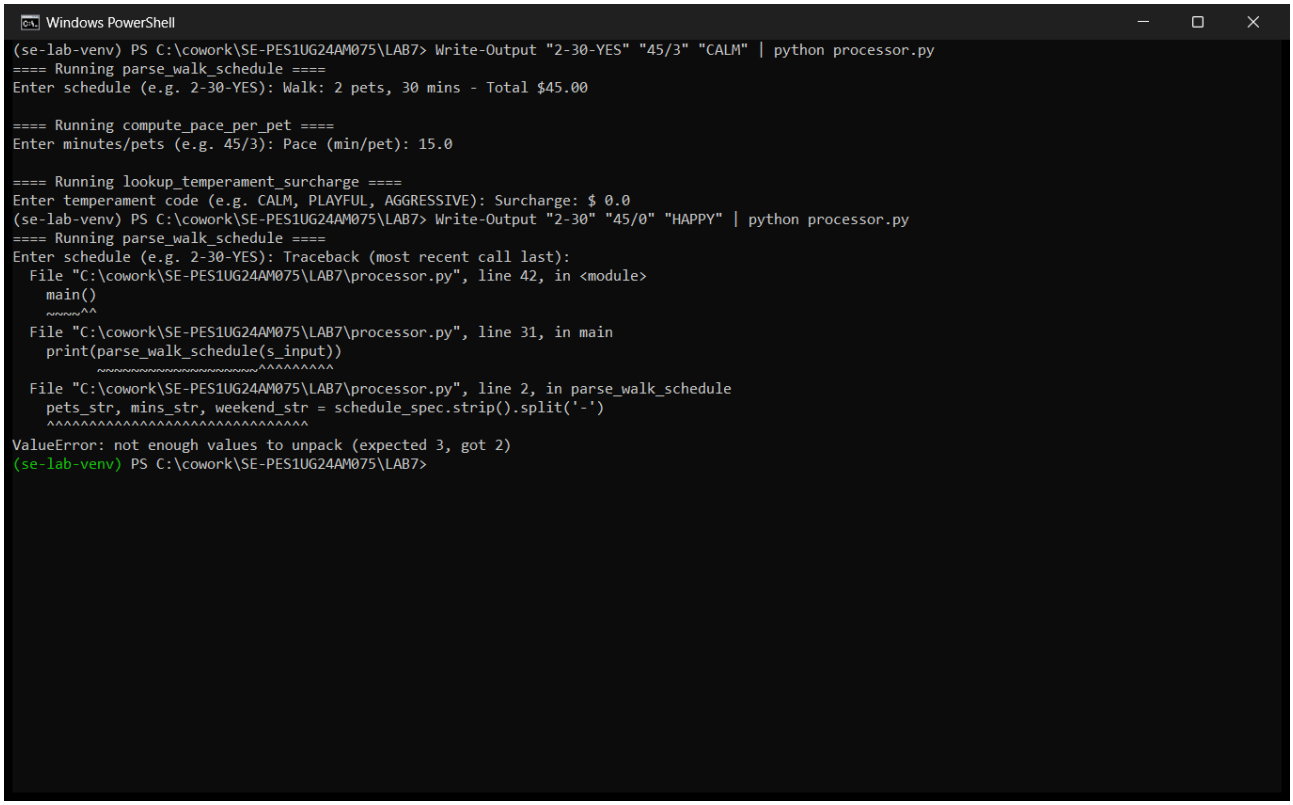
parse_walk_schedule
Bug: Crashed on None, missing or extra '-' parts, non-numbers, unknown weekend values and over-long digit strings; accepted 0 or negative pets and minutes.
Fix: Return None unless there are exactly 3 parts, pets is a whole number from 1 to 20, minutes from 1 to 480, and the weekend part is YES or NO.

compute_pace_per_pet
Bug: Crashed on None, missing '/', non-numbers and 0 pets; returned nan, infinity or negative paces.
Fix: Return None unless minutes is a finite number from above 0 to 480 and pets is a whole number from 1 to 20.

lookup_temperament_surcharge
Bug: Crashed on None and on any code without a known prefix, such as '' or 'HAPPY'.
Fix: Return None for non-text input and unknown codes.

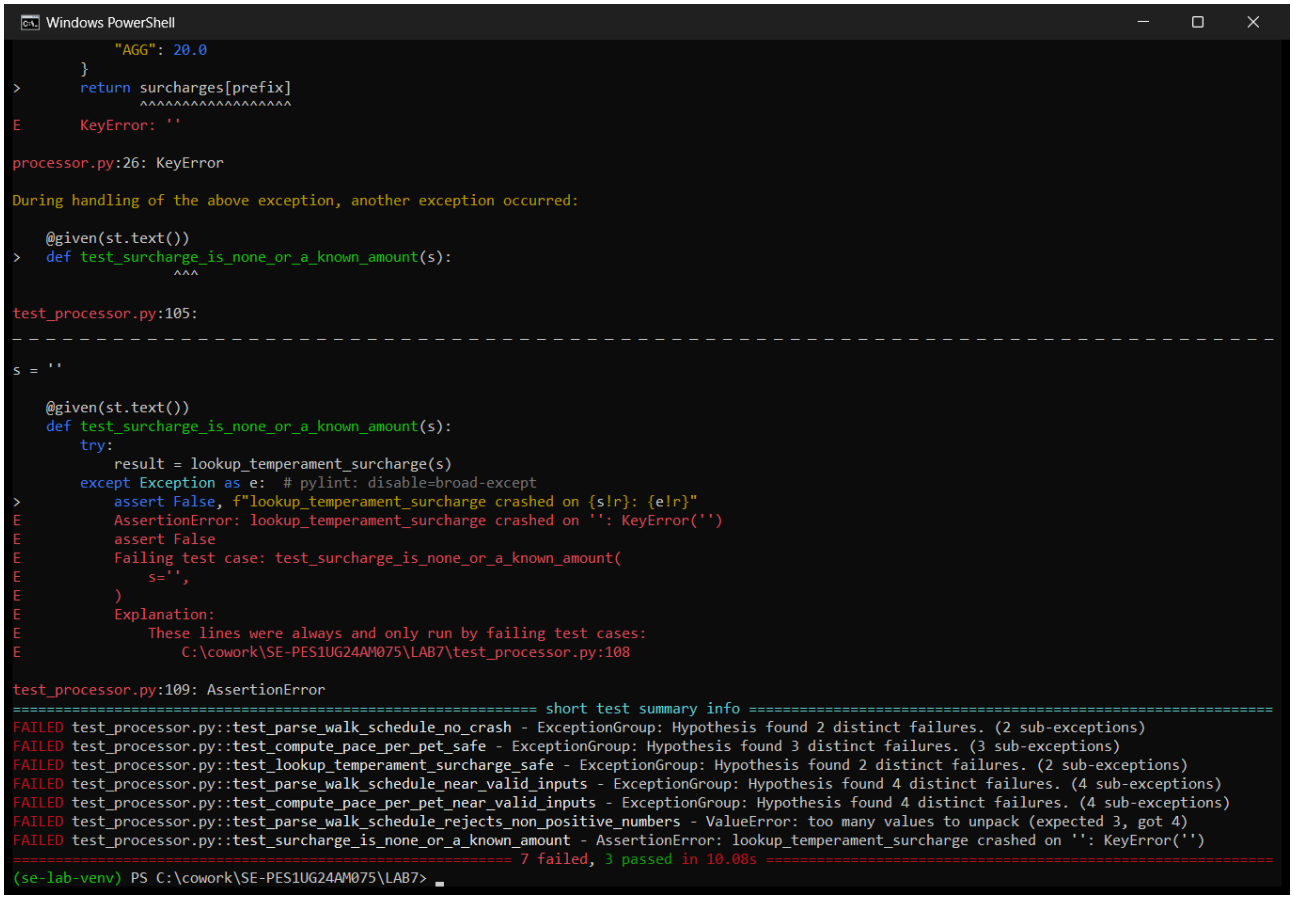
Step 2: Run the buggy processor.py

Command: python processor.py. Valid input works, but the schedule **2-30** (missing the weekend part) crashes with ValueError.



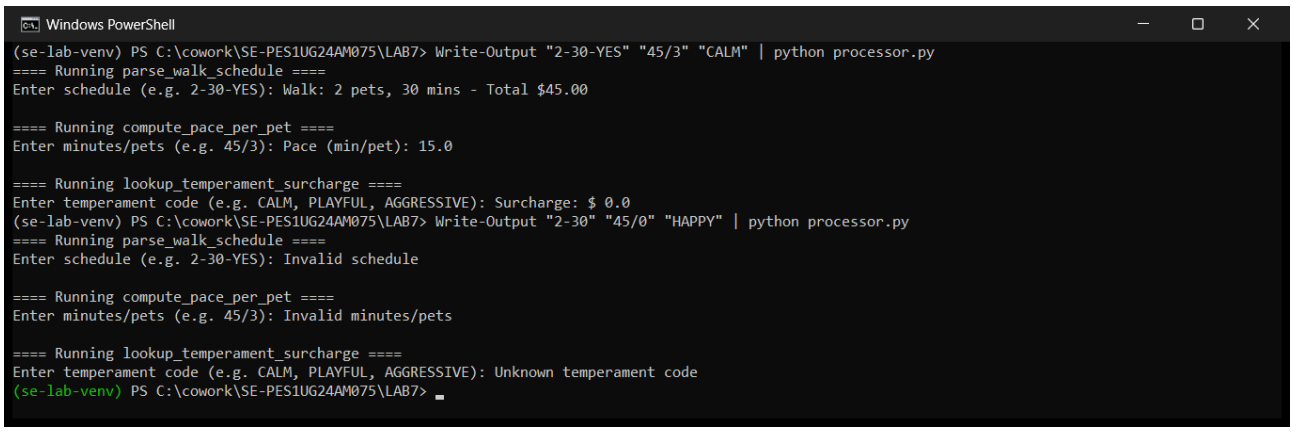
Step 4: Run test_processor.py against the buggy code

Command: python -m pytest test_processor.py. Hypothesis finds crashes in all three functions: 7 tests fail, 3 pass.



Step 5: Run the fixed processor.py

Command: python processor.py. Valid input gives the same results as before; invalid input now prints a message instead of crashing.



Step 6: Re-run test_processor.py against the fixed code

Command: python -m pytest test_processor.py. All 10 tests pass.

